Analysis of Firefly Synchronisation Model

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Mathematical Modelling I (25MAP111)

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Abstract

The report investigates the dynamics of firefly synchrony, using the Kuramoto model for synchronisation. In this project, numerical simulations were used to study factors affecting firefly synchronisation, including coupling strength, the number of fireflies, inter-firefly similarity, and the effect of noise. Clustering was also explored in the paper, showing the impact of how fireflies may synchronise in clusters rather than the system as a whole.

The results showed that increasing coupling strength rapidly increases synchronisation past the critical coupling, whilst expanding the number of fireflies decreases the time taken and allows for fewer fluctuations on the way to synchronisation. When natural frequencies are similar, we see faster synchronisation. When white Gaussian noise is included, we observe fluctuations that increase with noise strength.

The Kuramoto model captures the features of firefly synchronisation, including clustering and sensitivity to noise. Despite the assumptions made, the model proved to be valuable in evaluating firefly synchronisation.

Key Words: Synchronisation, Phase Locking, Clustering, Noise, Kuramoto model, Coupling Strength

Contents

1 Introduction

Fireflies exhibit one of the most beautiful synchronisation conditions in nature. Thousands of male fireflies gather in trees at night so that they can synchronise their lights so that they can flash on and off in unison. Meanwhile, the female fireflies cruise overhead looking for males with a handsome light. How does the synchrony actually occur? Initially, fireflies are not synchronised; synchrony develops as the night progresses. The key to this synchrony lies in the fireflies' mutual influence. When one firefly sees the flash of another, it will adjust its flash accordingly to flash more in phase on the next cycle. [1, 2]

We can leverage mathematical models to provide a systematic way to describe and analyse the phenomenon of firefly synchronisation, thereby understanding how it occurs in a group of fireflies. In this paper, a model is developed for a group of fireflies gathering in a tree, synchronising to find a female mate.

The model we analyse is the Kuramoto model, which is a mathematical model for the behaviour of a large set of coupled oscillators. This model will help us understand the conditions under which fireflies synchronise in a system, making it a valuable tool for explaining how the synchronisation occurs for a group of fireflies. [3]

This report focuses on a tree where male fireflies gather to flash their lights on and off in unison to attract female fireflies who are looking for a handsome light. Our goal is to evaluate a model of this synchronisation to understand and explain this phenomenon.

2 Background

2.1 Synchronisation

Synchronisation is the coordinated evolution of two or more oscillators due to interactions or couplings between them. The adjustment can be described in terms of phase locking and frequency entrainment. Synchronisation properties of self-sustained oscillators are based on the existence of phase ϕ. Phase is the variable parameterising the motion of an oscillator in a cycle. [4] [5] We can choose a phase in a way that it grows uniformly with time :

$$\frac{d\phi}{dt} = \omega_0 \tag{1}$$

where t is time, ω_0 is the natural frequency of oscillations. We can generalise a weak interaction affecting only the phases of two oscillators ϕ_1 and ϕ_2, by expanding on Equation 1 we now have:

$$\frac{d\phi_1}{dt} = \omega_1 + \varepsilon Q_1(\phi_1, \phi_2), \qquad \frac{d\phi_2}{dt} = \omega_2 + \varepsilon Q_2(\phi_2, \phi_1) \tag{2}$$

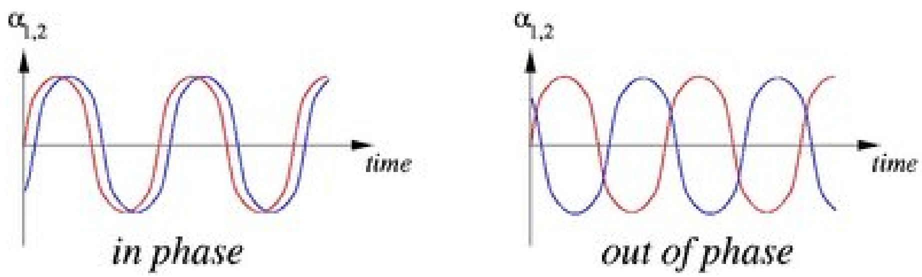

Figure 1: Synchronisation states for two coupled oscillators, which are nearly in-phase on the left and completely out of phase on the right, ($\alpha_{1,2}$ represent an oscillatory observable, e.g a deviation of a pendulum for coupled clocks in the Huygens experiment.

Source: `http://www.scholarpedia.org/article/Synchronization`

2.2 Clustering of Oscillators

Clustering occurs when groups of similar oscillators synchronise their behaviour internally, forming stable clusters. Potentially being desynchronised from other clusters in the network [6]. The clustering involves the following stages: (i) oscillators have their own phase running independently, starting from their own initial conditions. (ii) As time evolves, objects with the highest similarity synchronise with each other first. (iii) afterwards, in a sequential process, more and more objects synchronise together, and clusters are produced (iv), finally, the population is split into several stable, distinct structures [7]

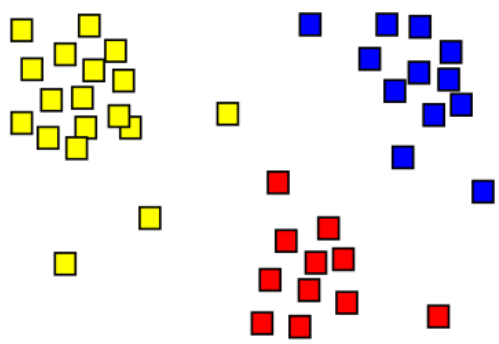

Figure 2: The result of clusters shown as coloured squares, which are split into three clusters

Source: `https://upload.wikimedia.org/wikipedia/commons/c/c8/Cluster-2.svg`

2.3 White Gaussian Noise

Real systems experience some noise, which affects synchronisation. Noise can account for unexpected phenomena that delay the synchronisation of all oscillators in a system; this can include confusion in fireflies or disruptions in a network. Even slight noise can cause a system to perform one more cycle before synchronising.

White noise is a commonly used random process. It is used to model the external noise in synchronisation systems. A stochastic process is called white noise if the power spectral density is constant for all frequencies. We can also consider a white Gaussian noise process, with zero mean and a Gaussian distribution. [8]

3 Model Description

3.1 Model Assumptions

This model considers a tree of male fireflies flashing to attract female fireflies. Using a single governing equation, we can simulate a range of fireflies. The following assumptions have been made in the model

- There is weak coupling between fireflies in the system
- Firefly oscillators are identical or nearly identical, except for their natural frequencies, which are drawn from the same distribution
- Interactions depend sinusoidally on the phase difference between each pair of fireflies
- The system will initially be considered without noise or random disturbances to the oscillators
- There are no time delays in how fireflies perceive and respond to flashes from other fireflies

These assumptions simplify the system and enable analysis of how male fireflies synchronise to attract female fireflies.

The model evaluated is the Kuramoto synchronisation model applied to fireflies synchronising with each other.

3.2 Dimensional Model

In this section, we define the model we use. This is the Kuramato synchronisation model for fireflies [3]s:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{1}{N}\sum_{j=1}^{N} K_{ij}\sin(\theta_j - \theta_i), \quad i = 1\ldots N \tag{3}$$

where the system is comprised of N oscillators, with phases θ_i and coupling strength K. Each oscillator is considered to have its own intrinsic frequency ω_i

We may also reduce our equation 3 to the mean field form, where:

$$K_{ij} = K, \quad \forall i, j \in [0, N] \tag{4}$$

With this reduction, we can change our equation from 3: to

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N} \sin(\theta_j - \theta_i), \quad i = 1\ldots N \tag{5}$$

With this reduction, we now have new assumptions:

- All-to-all coupling: all fireflies can observe and influence each other equally, regardless of distance
- Coupling is symmetric so that if firefly i influences firefly j with some strength then firefly j influences firefly i with the same strength

3.3 Order Parameter

To measure the synchronisation among fireflies in the Kuramoto model, we can introduce the order [9, 10]:

$$re^{i\psi} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j}, \quad r \in [0, 1] \tag{6}$$

Where r represents the phase coherence of the population of oscillators and ψ indicates the average phase.

3.4 Critical Coupling

We should also take into account the critical coupling, which is where the phase transition from incoherency to synchronisation will occur [11]:

$$K_c = \frac{2}{\pi g(0)} \tag{7}$$

where g is a symmetric, continuous, and unimodal distribution centred above 0

3.5 Introducing Noise to Model

We may also introduce noise to our system in 5, so that we alter it to:

$$\frac{d\theta_i}{dt} = \omega_i + \zeta_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \quad i = 1 \dots N \tag{8}$$

where ζ_i is the fluctuation and a function of time. We will also consider the noise to be a Gaussian white noise so that:

$$\begin{aligned} &\langle \zeta_i(t) \rangle = 0, \\ &\langle \zeta_i(t)\zeta_j(t') \rangle = 2D\delta_{ij}\delta(t - t') \end{aligned} \tag{9}$$

with D denoting the strength of the noise. Noise accounts for external factors such as environmental disturbances and imperfect perception of other fireflies [3].

3.6 Non-Dimensional Model

In most cases, to simplify analysis and reduce the number of parameters, we can introduce a dimensionless model with dimensionless parameters. However, in the case of our model, Equation 3, non-dimensionalising didn't reduce the number of effective parameters. We can non-dimensionalise our model from Equation 3, by defining dimensionless parameters:

$$t^* := \frac{t}{\tau}, \quad \tilde{\omega_i} := \tau\omega_i, \quad \tilde{K_{ij}} := \tau K_{ij} \tag{10}$$

With these dimensionless variables, we get our dimensionless model:

$$\frac{d\theta_i}{dt^*} = \tilde{\omega} + \frac{1}{N}\sum_{j=1}^{N} \tilde{K_{ij}} \sin(\theta_j - \theta_i), \quad i = 1 \dots N \tag{11}$$

Although we now have a dimensionless model, we have not significantly reduced the number of effective parameters. For this reason, we will retain the dimensional form of the model for the subsequent analysis.

4 Results and Discussion

4.1 Uncoupled Fireflies

Let's consider fireflies that do not wish to synchronise with each other. To do this, we can adjust our coupling constant so that:

$$K = 0$$

Now we have fireflies who do not wish to synchronise with each other. This gives us a new equation in the form of:

$$\frac{d\theta_i}{dt} = \omega_i \tag{12}$$

As we can see, we now have an equation for a firefly which is unaffected by the coupling effects of other fireflies, or a firefly which is unaffected, as in this case, fireflies do not influence each other.

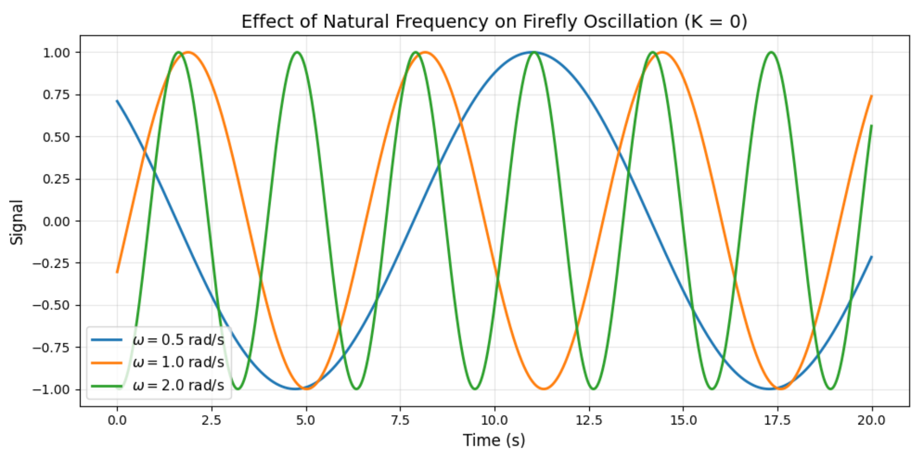

Figure 3: 3 Unsynchronised Fireflies with varying natural frequencies starting from randomised points where $A = \sin \omega t$

In Figure 3, we observe that the three male fireflies exhibit sustained natural behaviour, with the oscillatory period determined solely by their natural frequencies, as expected from Equation 12. We also observe no sign of synchronisation due to that standard behaviour, indicating that the fireflies are unaffected by one another in the case of no coupling constant K.

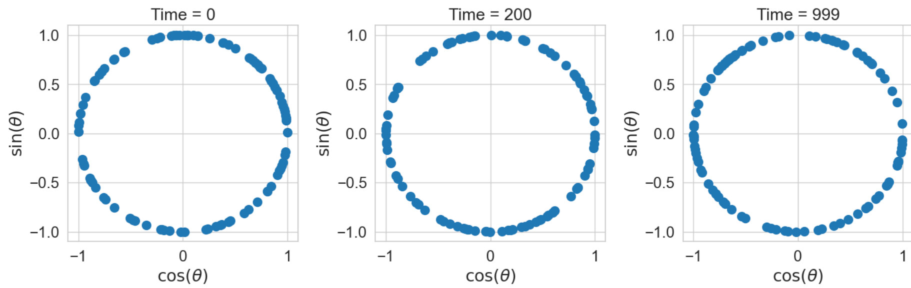

Figure 4: Evolution of individual fireflies with randomised natural frequencies who are not synchronising

By plotting 100 fireflies who do not want to synchronise in a complex plane in Figure 4, we can see that although the fireflies fluctuate as time goes on, they do not synchronise and show no signs of synchronising due to the fireflies being unaffected by each other.

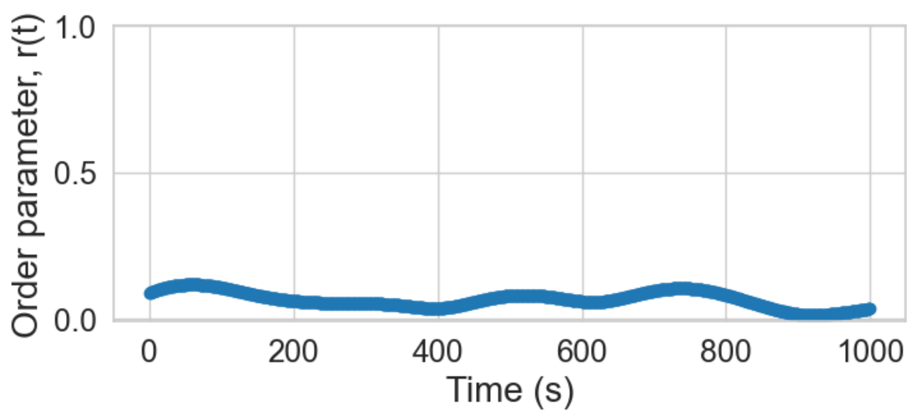

Figure 5: Order parameter variation over time for fireflies that are not coupling with each other. Where the degree of synchronisation is quantified using the order parameter in Equation 6

By using the same randomised fireflies as in Figure 4, we can see that by plotting the order parameter in Figure 5, the order parameter stays close to zero at all times, as there is no synchronisation occurring. There are small fluctuations arising from random phase alignments among the fireflies; these alignments are also finite, the fireflies eventually lose their alignment, as seen from the fluctuations caused by the absence of a coupling constant K.

From these, we see that fireflies with no coupling constant act as independent oscillators, showing no signs of synchronisation over time. We can use this section as a baseline for the effects of synchronisation in the rest of the section, where we will now have a coupling constant K.

4.2 Coupling Fireflies

Let's consider fireflies that do wish to synchronise with each other. To do this, we can adjust our coupling constant so that:

$$K > 0$$

To demonstrate clear synchronisation within the firefly system, we will set the coupling constant so that:

$$K = 3$$

Now we have fireflies who wish to synchronise with each other as they are affected by the coupling constant. We can now make use of our Equation 5.

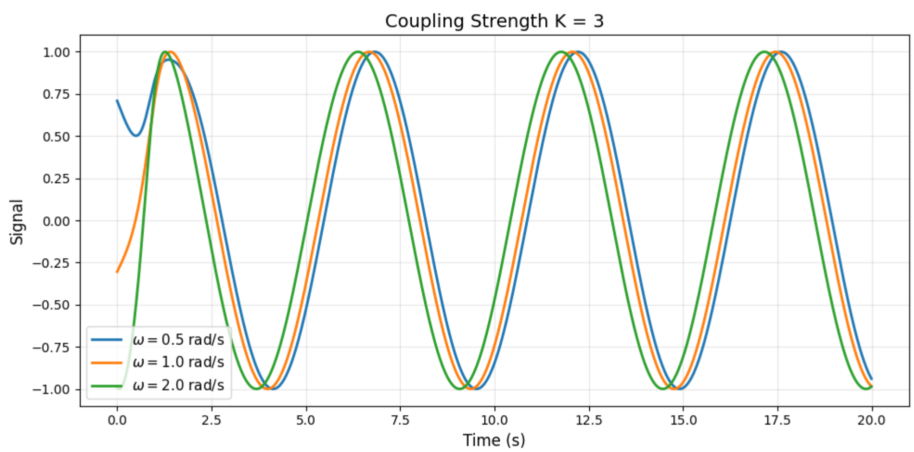

Figure 6: 3 Synchronised Fireflies with varying natural frequencies starting from the same randomised points as Figure 3 where $A = sin\omega t$ and $K = 3$

In Figure 6, we observe that the three male fireflies synchronise, with the oscillatory period determined by each other as there is now a strong coupling constant, as expected from Equation 5.

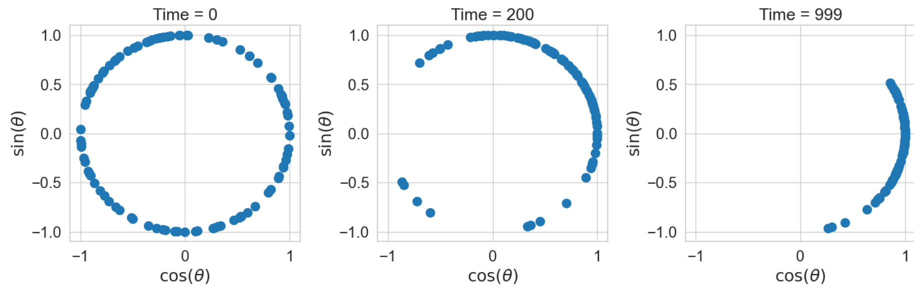

Figure 7: Evolution of individual fireflies with randomised natural frequencies who are synchronising

By plotting 100 fireflies who want to synchronise in a complex plane in Figure 7, we see that as time goes on, the fireflies group together, showing that they are synchronising, and as time goes on, we can observe a singular firefly

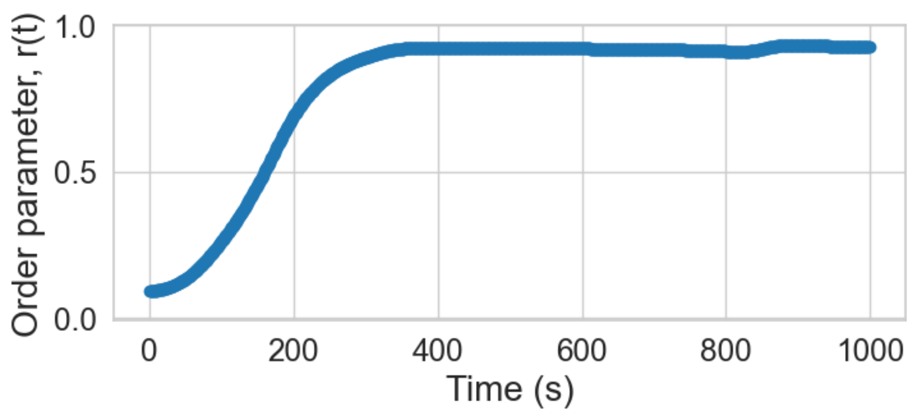

Figure 8: Order parameter variation over time for fireflies that are coupling with each other. Where the degree of synchronisation is quantified using the order parameter in 6

By using the same randomised fireflies as in Figure 7, we can see that by plotting the order parameter in Figure 5, the order parameter stays close to one as there is synchronisation occurring. In contrast to Figure 5, there are almost no fluctuations, showing that once the fireflies are more synchronised, they stay synchronised, caused by the presence of a coupling constant K.

From these, we can see that fireflies with a coupling constant can synchronise, as there will be signs of synchronisation as time goes on. In contrast to the graphs in Subsection 4.1, where there was no coupling constant.

4.2.1 Effect of Coupling Strength

The coupling strength K determines how strongly a firefly affects another. We can vary the coupling strength to observe effects at different values of K to see the impact of increasing K on Equation 5.

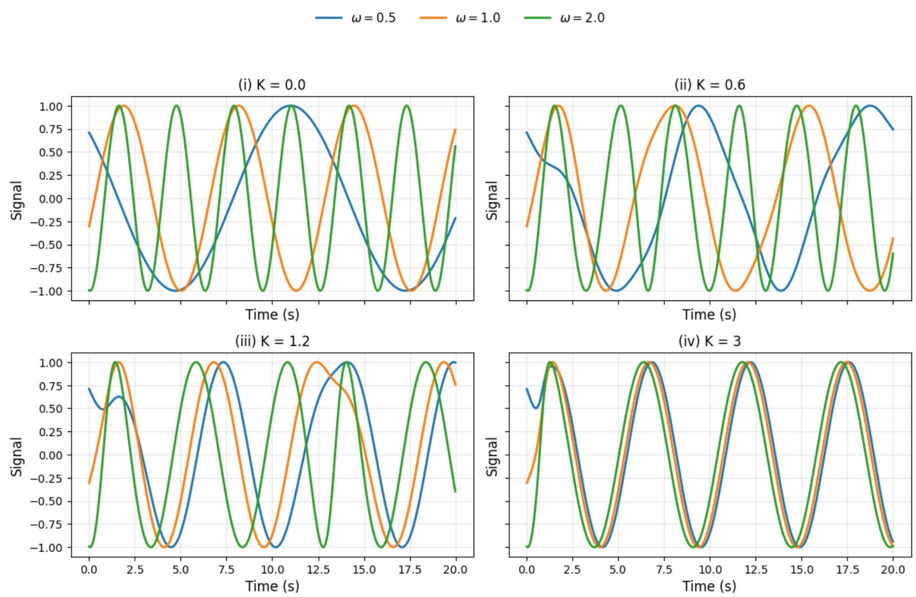

Figure 9: Graphs with varying coupling strengths K where there is the same randomised start and $N = 3, (\omega_1, \omega_2, \omega_3) = (0.5, 1, 2)$

From Figure 9, we see that as we increase the coupling constant K, the fireflies synchronise more quickly and are overall more synchronised. We see that the graph (i) resembles Figure 3, as there is no coupling constant; the fireflies are independent and unaffected by each other. Since there is no coupling strength K in this case, the equation becomes Equation 12. In (ii), we can see that there are apparent changes in the firefly flashing, and now we do not have simple oscillations. In (iii), we can see that the firefly flashing is more closely synchronised than in (ii), as the firefly has more similar flashing due to a stronger coupling strength. In (iv), we can see that the fireflies are more significantly synchronised as they almost instantly synchronise, then observe similar flashing due to the considerably higher coupling strength K than in (iii)

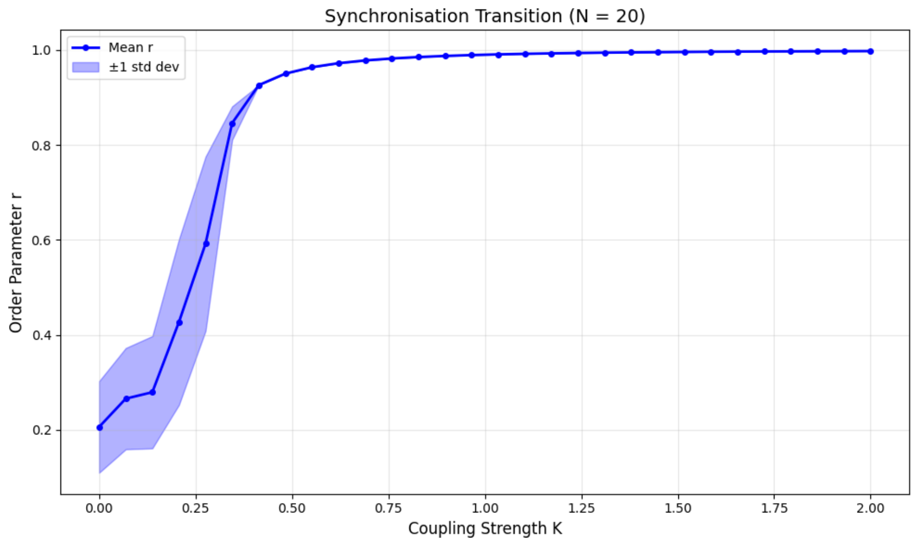

Figure 10: Synchronisation curve for increasing coupling strength K. We used $N = 20$ fireflies with randomised frequencies. There is also a critical coupling of $K_c = 0.276$. Each test was simulated for 50 seconds

From Figure 10, we can see that as we increase coupling strength, synchronisation increases, as seen through the increasing order parameter. The graph begins at $K = 0$; we see that the order parameter does not start at 0; this is because, initially, some synchronisation occurs by chance. At this point of the graph, we can consider the fireflies as simple oscillators modelled by Equation 12.

We can also see a sharp rise in $0.2 < K < 0.3$, which is due to the critical coupling K_c being within this area. We can see that when $K < K_c$, there is a significantly weak coupling between the fireflies. Then, when we get to $K = K_c$, there is a sharp rise in the order parameter, bringing it to 0.848. After $K > K_c$, there is strong coupling as the order parameter gets to a value close to 1.

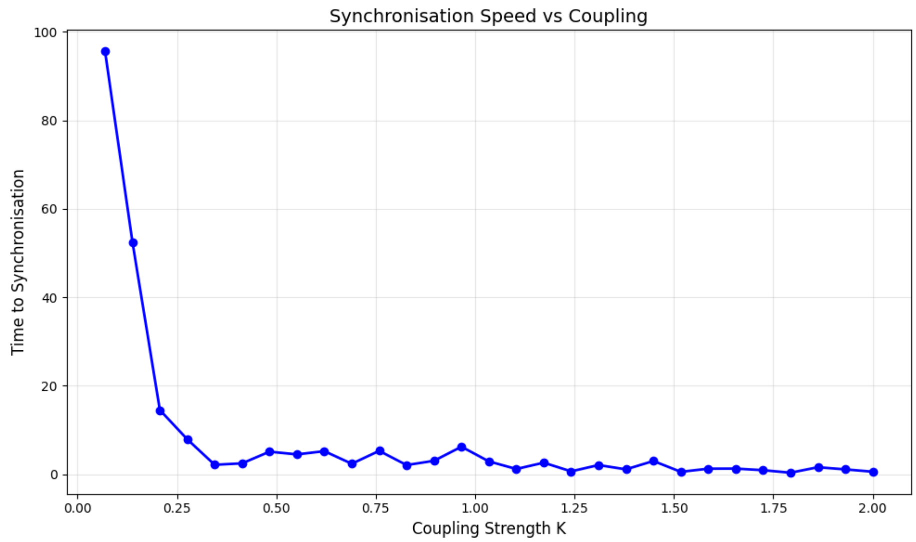

Figure 11: Time curve showing how the effects of the coupling strength K affect how long the fireflies take to synchronise. We used N = 20 fireflies with the same randomised frequencies from Figure 10. Synchronisation was registered when the order parameter got above 0.95

From Figure 11, we can see that as we increase coupling strength, the time it takes to synchronise decreases, as seen from the decreasing y value. As we can see in contrast to Figure 10, our coupling constant does not start at 0; this is because if 0, the fireflies would not synchronise, which would lead to infinite time, for that reason, we will not consider the simple oscillator fireflies modelled by Equation 12.

In contrast to Figure 10. We can see a sharp drop in $0.2 < K < 0.3$, where we would find the critical coupling K_c. We can see that when $K < K_c$, the fireflies would take a long time to synchronise. Then, when $K = K_c$, there is a sharp drop in how long it takes for the fireflies to synchronise. Afterwards, when $K > K_c$ we can see that it does not take long at all for the fireflies to synchronise.

4.2.2 Effect of Number of Oscillators

The number of fireflies N determines how many fireflies are taking part in the system. We can vary the number of fireflies to observe the effects at different values of N to see the impact of increasing N on Equation 5

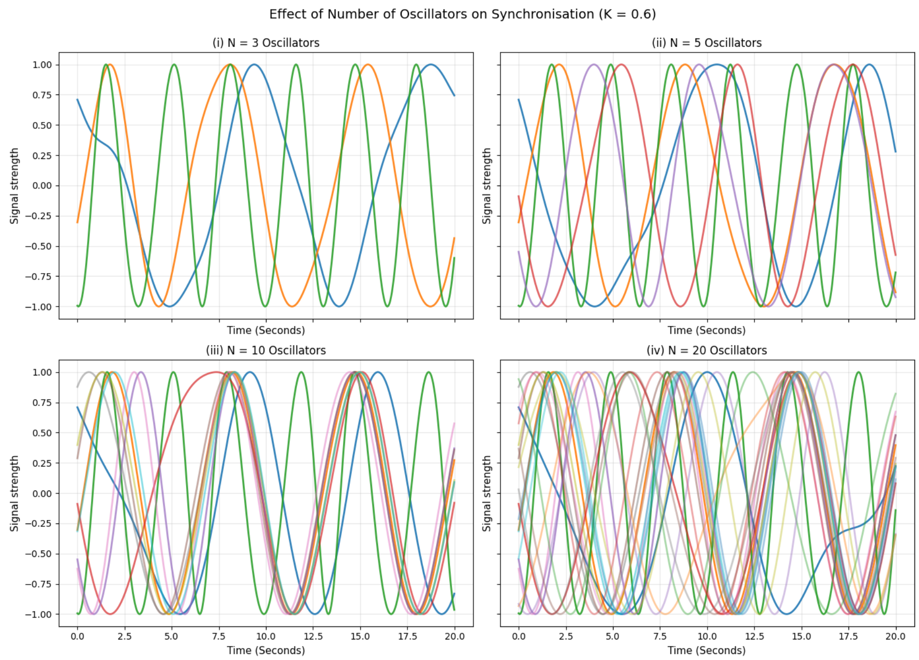

Figure 12: Graphs with varying firefly amounts N where there are randomised starts and K = 0.6. The first three fireflies are as in Figure 9, after which we get new random fireflies, which we cumulatively add on as we increase our number of fireflies.

From Figure 12, we see that increasing the firefly amount N results in more apparent synchronisation. We see that on graph (i), there is not much evidence of the fireflies synchronising with each other. Looking closely at (ii), we can see that the introduction of 2 new fireflies allows the fireflies to sync up more than before. In (iii), where the amount of fireflies is doubled, we see significantly more synchronisation. Afterwards, in (iv), when we double the number of fireflies again, we observe much more synchronisation than before. We can assume that this is because, as there are more fireflies, a firefly will experience an average over more neighbours, meaning outlier fireflies will contribute less to the mean field.

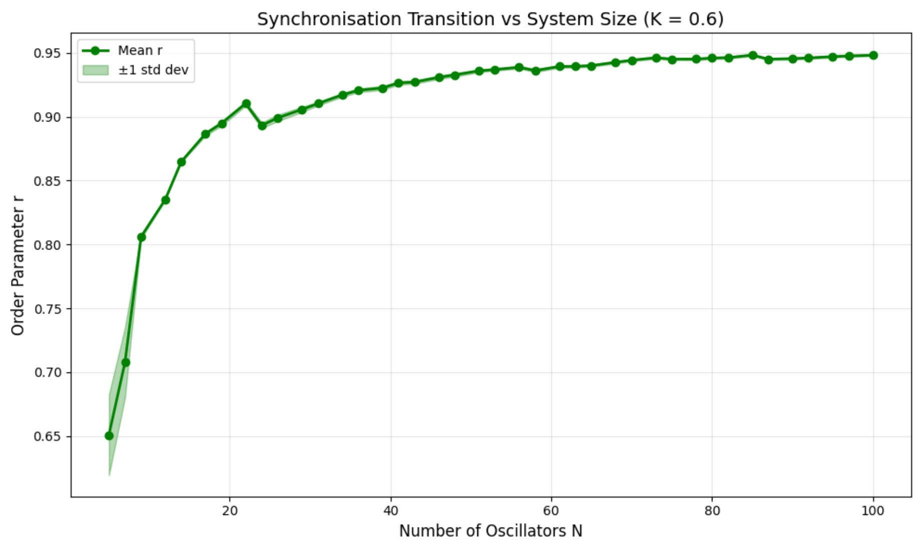

Figure 13: Synchronisation curve for increasing fireflies N. We had fixed $K = 0.6$ with randomised frequencies. Each number of oscillators was simulated for 50 seconds

From Figure 13, we can see that as we increase the number of fireflies in the system, synchronisation rises and becomes more predictable and much less liable to random fluctuations, which is due to the firefly experiencing more neighbours, as each oscillator will end up making up a smaller amount of the mean field. We also observe that, for $30 < N < 40$, r changes very little with increasing. This graph further demonstrates that more oscillators allow greater coherence.

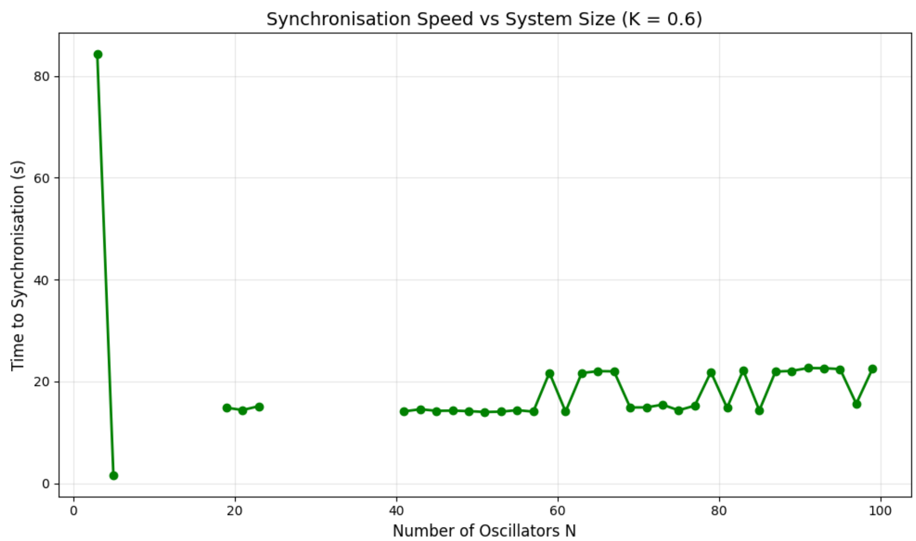

Figure 14: Time curve showing the effects of how the number of fireflies N affects how long it takes for fireflies to synchronise. We used $K = 0.6$ with randomised frequencies for the fireflies. Synchronisation was observed when the order parameter exceeded 0.9. The simulations had run for 100 seconds maximum

From Figure 14, we can see that increasing the number of fireflies does not necessarily mean that there is a faster synchronisation. However, this means there is a greater likelihood that synchronisation will occur, as the impact of an outlier firefly is reduced. We also observe cases in which the fireflies do not even synchronise within the allotted time.

4.2.3 Effect of Frequency Distribution Width

We can use σ to denote the variance of the natural frequencies of the fireflies. We can vary the variance to see how different values of σ affect firefly synchronisation.

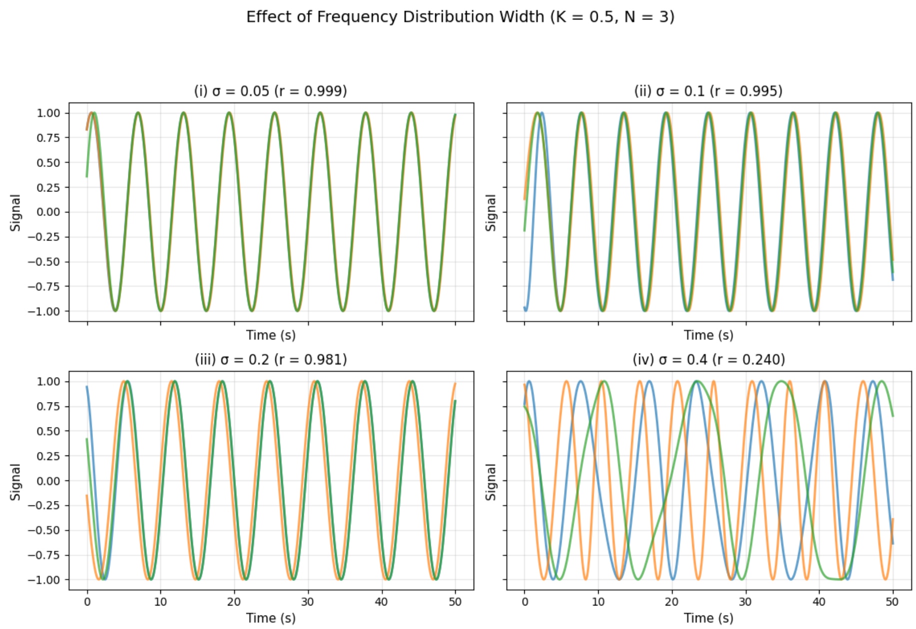

Figure 15: Graphs with different variances where coupling strength is $K = 0.5$ and there are $N = 3$ fireflies. The firefly's natural frequency is randomly drawn from a normal distribution, with variance determined by the variance σ

From Figure 15, we see that increasing the variance σ results in less synchronisation between the fireflies, suggesting that the variance of frequencies decreases firefly coherence. We see that on graph (i), the fireflies are more coherent, so much so that they appear as a single line, as indicated by the high order parameter, which is almost perfect synchronisation. On (ii), we can see a nearly identical line to (i) that is slightly thicker. On (iii), we can now more clearly see the other fireflies, but they are still quite coherent. On (iv), we observe that the fireflies are highly unsynchronised, as confirmed by the low order parameter.

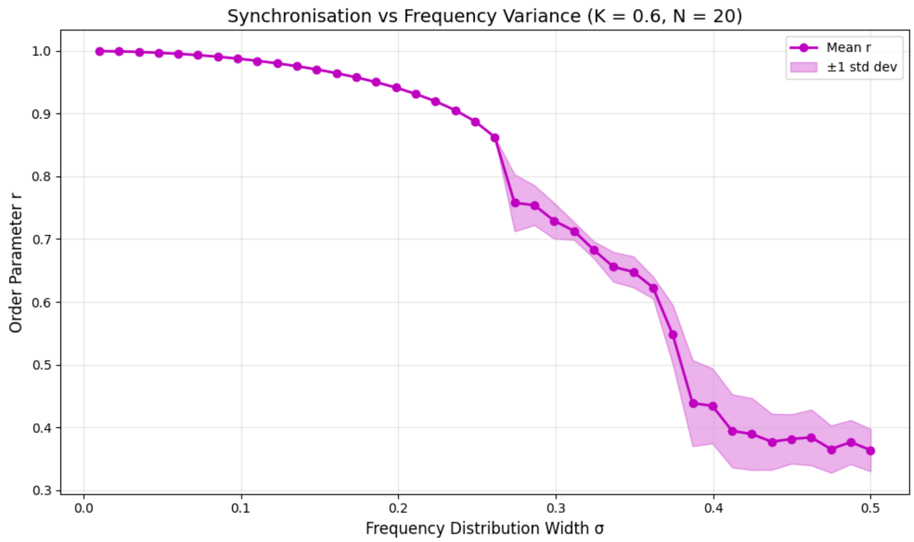

Figure 16: Synchronisation curve for increasing σ. We had fixed $K = 0.6, N - 20$ with randomised frequencies within the distribution width. Each distribution width was run for 50 seconds

From Figure 16, we see that as we increase the width of the frequency distribution of the fireflies, the order parameter decreases additionally, when the increasing frequency distribution width shows much less predictable behaviour. When there is a low σ, fireflies will have similar natural frequencies, making it easy for them to synchronise with each other, When there is a high σ, we have fireflies with quite different natural frequencies making it harder for the fireflies to synchronise to each other

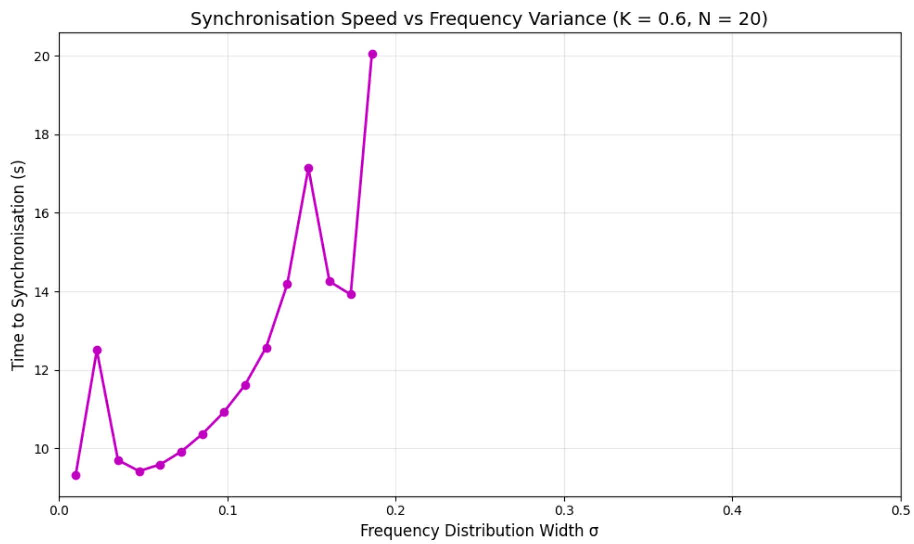

Figure 17: Time curve for increasing σ. We fixed $K = 0.6$ and $N = 20$ with randomised frequencies within the distribution's width. Each distribution width was simulated for 50 seconds. Synchronisation was observed when the order parameter exceeded 0.9. The simulations ran for at most 600 seconds.

From Figure 17, we can see that when increasing the distribution width σ, fireflies will take longer to synchronise. We also observe greater fluctuations in how long synchronisation will take. This is because it is harder for fireflies to synchronise with each other when they have significantly different natural frequencies. We observe that when $\sigma > 0.15$ we observe an incredibly sharp increase in the time taken to synchronise, as shown by $\sigma > 1.75$, where we have fireflies not synchronising within the 600-second simulations.

4.2.4 Effect of Noise

Here we will consider noise ζ_i in the Kuramoto model, Equation 8. By introducing white Gaussian noise, we will show how it affects firefly synchronisation. In the case of firefly synchronisation, we can interpret noise as a firefly not having the best gauge of another firefly's frequency or bad weather, which may affect a firefly's ability to synchronise

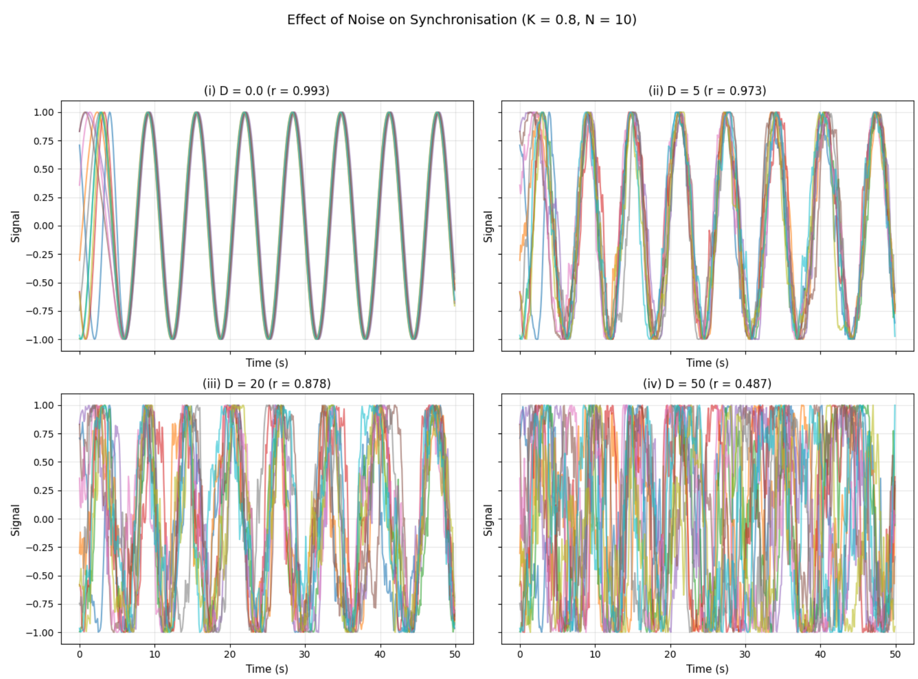

Figure 18: Graphs with varying levels of noise D, where coupling strength is $K = 0.8$ and there are $N = 10$ fireflies. The fireflies' natural frequencies are randomly drawn from a normal distribution at the start and are consistent throughout the graphs. R is the order parameter, which is averaged from the last 20% of the simulation to account for random fluctuations.

From Figure 18, we can see that as we increase the noise, we can see more fuzzy lines. At (ii) is not as synchronised, and we can see that we still observe some phase locking but with fluctuations, which still gives a high order parameter. Afterwards, at (iii), we can see that there is a significant increase, and the oscillators are not as tightly synchronised as before. Finally, in (iv) it is hard to make out what is happening at all, but there may still be some random synchronisation among the fireflies, as the order parameter is not close to 0. We also observe that although noise affects the system, it must be increased by a considerable amount to have a significant impact. We can also see oscillators fluctuating around the mean phase even when synchronised.

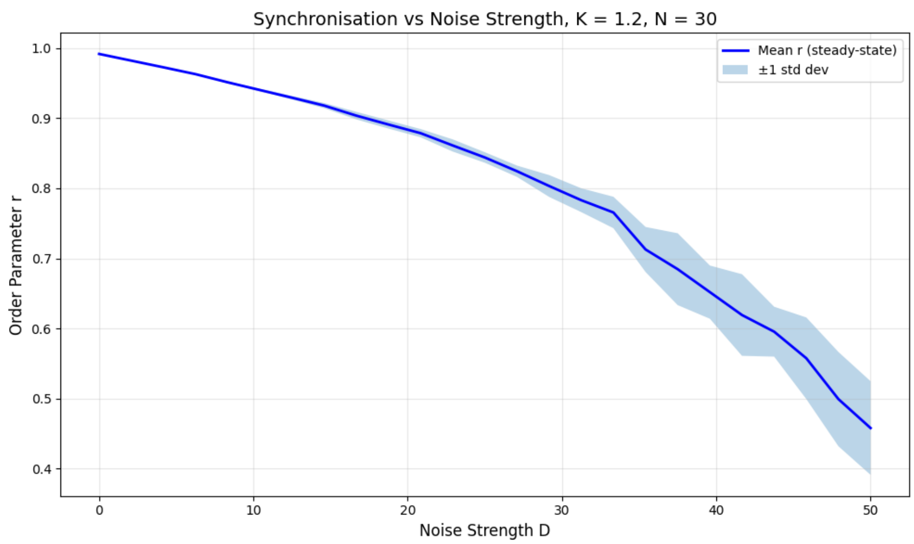

Figure 19: Synchronisation curve for increasing D. We had fixed $K = 1.2, N = 30$, with randomised frequencies within the distribution width. 15 trials were run for each D tested, and simulations were capped at 300 seconds.

From Figure 19, we can see that as we increase the noise D, we tend to get a decrease in the order parameter, but we can also see from that the standard deviation increases, implying that increased noise makes it so that the system becomes more unpredictable. We can also see that noise does not entirely shut off synchronisation.

4.2.5 Summary of Parameter Effects on Synchronisation

Parameter	Increasing	Decreasing
Coupling Constant, K	r increases and faster synchronisation once $K > K_c$	r fluctuates near 0 and slower synchronisation
Oscillator Amount, N	r increases, less prone to fluctuations	r decreases, more prone to fluctuations
Frequency Distribution Width, σ	r decreases and more prone to fluctuations	r increase and less prone to fluctuations
External Noise, D	r decreases, and more prone to fluctuations	r increases and less prone to fluctuations

Table 1: A summary table breaking down the effects of varying the parameters of the Kuramoto model on firefly synchronisation

4.3 Clustering Fireflies

In contrast to full synchronisation, in which all fireflies synchronise with each other, clustering is a phenomenon in which fireflies synchronise within their groups rather than the system as a whole initially. This occurs when fireflies initially synchronise with neighbours who have similar natural frequencies, as it is easier to sync with those that have comparable natural frequencies than those that do not. In this section, we will explore clustering.

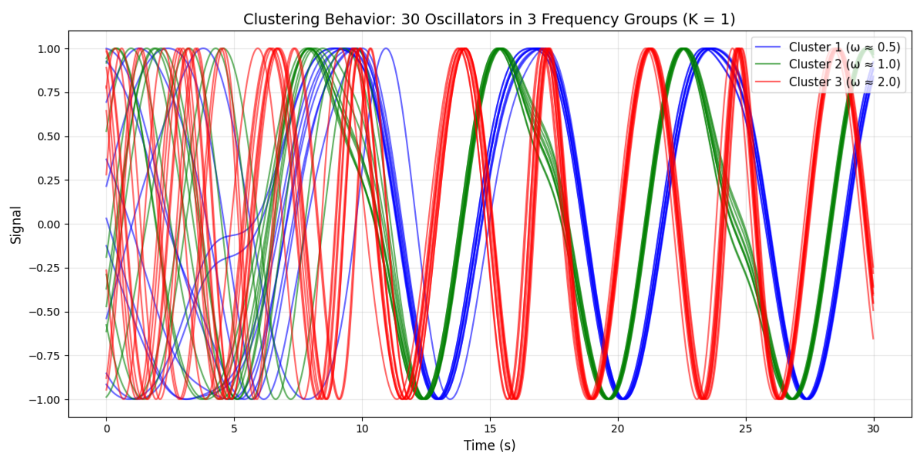

Figure 20: Fireflies synchronising within their clusters of 10 fireflies where $N = 30, K = 1$, here frequencies were randomly generated within $\omega - 0.05 < \omega < \omega + 0.05$ range so that we have similar firefly frequencies

From Figure 20, we can see that as time goes on, the fireflies will synchronise with those that have similar natural frequencies in clusters, which results in 3 different waves going on with different frequencies.

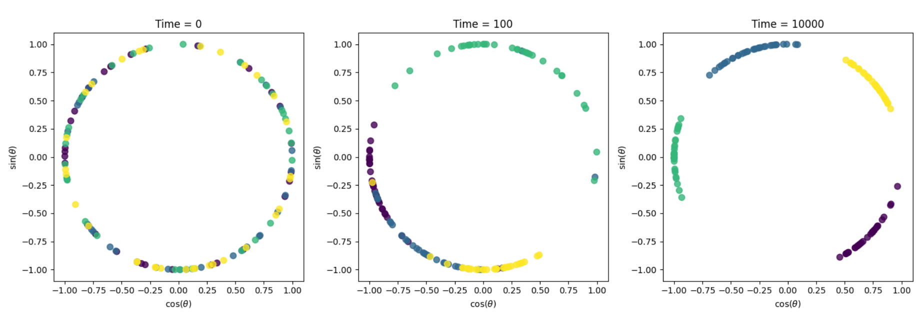

Figure 21: Evolution of individual fireflies with randomised natural frequencies who are synchronising in clusters, where each colour demotes its own cluster

From Figure 21, we can see that in plotting 128 fireflies who wish to synchronise in a complex plane, as time goes on, there is synchronisation, but in clusters rather than with each other in clear, distinct areas of the graph at $Time = 10000$.

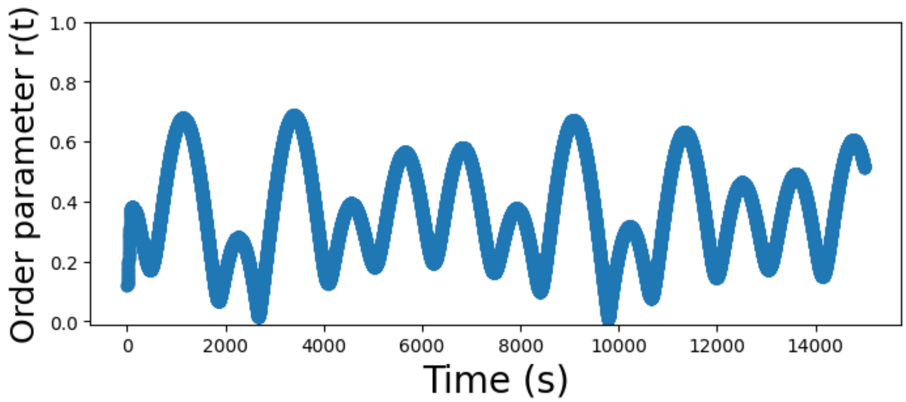

Figure 22: Order parameter variation over time for fireflies that are coupling in clusters.

By using the same randomised fireflies as in Figure 21, we can see that by graphing our order parameter against time. In contrast to Figure 4, we have larger peaks as there is apparent synchronisation occurring. The dips arise from clusters that counteract one another, thereby reducing the order parameter.

5 Conclusion

The study analysed the Kuramoto Synchronisation Model and explored the effect of adding white noise on firefly synchronisation. The results demonstrated the impact that coupling strength K, firefly amount N, variance of frequencies σ, and noise strength D have on the fireflies synchronising.

In the absence of noise, we were able to see how the coupling strength affects the rate at which fireflies synchronise to each other, especially past the critical coupling point. We also observed that increasing the number of fireflies in the system makes it less prone to fluctuations during synchronisation. We also confirmed that fireflies synchronise better with those that have similar natural frequencies. In the presence of noise, we observed that a large amount of noise would be required to decrease the order parameter significantly.

We also observed clustering behaviour, showing that clustering differs from regular synchronisation. We observed graphically how different clusters may look and counteract each other to reduce the order parameter, even when clusters are synchronised.

Overall, the Kuramoto model allowed us to explore different themes in firefly synchronisation by modelling fireflies as coupled oscillators. We were also able to capture real-world scenarios in the model, including noise and clusters. Future works can consider other network topologies such as a ring topology which are cannot use the mean field form of the Kuramoto model, rather than the all-to-all topology explored in this paper.

A Acknowledgements

I want to thank the members of my group, Aditya Kottapalli and Giulia Conway, for helping me better understand the concept of synchronisation, as I have not worked in this much depth on the topic of Dynamic Systems. The regular meetings deepened my understanding of the topic, enabling me to grasp it as much as possible. I would also like to thank the lecturer, David Sibley, for his guidance in modelling, his suggestions, and his efforts to keep us on the right track.

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